

Broadband body-worn antenna using a top-loaded pyramidal radiator

ISSN 1751-8725

Received on 24th September 2019

Revised 5th November 2019

Accepted on 2nd January 2020

E-First on 10th February 2020

doi: 10.1049/iet-map.2019.0825

www.ietdl.org

Brad R. Jackson¹ ✉, Eric Merkley²¹Electrical & Computer Engineering, California State University, Northridge, CA, USA²Department of Electrical & Computer Engineering, Queen's University, Ontario, Canada

✉ E-mail: bradley.jackson@csun.edu

Abstract: A wide-bandwidth low-profile quasi-omnidirectional antenna is presented and experimentally demonstrated when integrated with a helmet for emergency services or military applications. The antenna consists of a pyramid-shaped radiator that is top-loaded with an etched copper-clad microwave laminate with two shorting strips used to improve the input impedance matching, particularly for lower frequencies. The design and operation of the antenna are first introduced using a flat ground plane, and then experimental results are reported when it is installed on a helmet where the outer surface of the helmet is used as the ground plane. The results show a very wide impedance bandwidth from 890 MHz to 13.7 GHz ($VSWR \leq 3$) and predominately monopole-like radiation patterns up to ~ 11 GHz. The antenna has a height of $\lambda/16$ at the lowest operating frequency of 890 MHz (21 mm). Simulation results with a human body model demonstrate the significantly improved omnidirectionality of the proposed helmet antenna compared to more conventional body-worn antennas used in some applications.

1 Introduction

In modern military and emergency services applications, the ability to transmit and receive wireless signals is essential, and in many cases this capability is safety-critical. Such applications generally require individuals to have agility and freedom-of-movement that is not impeded by the required supporting equipment, including the antenna(s) worn or carried.

For military applications, in particular, dismounted soldiers may need to conduct electronic warfare (EW) operations that require wide bandwidth antennas in order to sense the electromagnetic environment for electronic support measures and/or to transmit jamming signals for electronic countermeasures (ECM). These applications, as well as conventional wireless communication, require omnidirectional antennas, which have conventionally been implemented with monopole or dipole antennas located close to the body. A particularly life-critical application is ECM for radio-controlled improvised explosive devices (RCIEDs). For this application, backpacks are often worn containing the electronics and battery, and the antennas protrude behind and above the shoulder(s) and head.

The close proximity to the human body can significantly affect the omnidirectionality of an antenna's radiation pattern. If the antenna is elevated above the head to alleviate this problem, a limitation in the mobility and agility of the wearer results. Given the typically narrowband nature of wire monopoles and dipoles, multiple separate antennas are required in some cases, which adds to the carrier's burden. A potential solution to this problem, which is explored in this paper, is the use of a single broadband low-profile antenna that is installed on top of a helmet. Such an antenna location could improve omnidirectionality, and, if the installed antenna minimally affects the size, weight, and function of the helmet, it could be an ideal solution for military and emergency services applications where helmets are a necessity.

Wide bandwidth low-profile omnidirectional antennas have received a significant amount of attention in recent years. A variety of different design techniques have been investigated such as the use of dielectric resonators [1] and wideband electrically small printed monopoles [2]. A monopole-based antenna that extends vertically above the helmet is perhaps most suitable to minimise the impact of the human body. For a helmet-based application, it is

important that such designs achieve a very low profile, and minimising the height of wideband monopoles has been the subject of a number of investigations (e.g. [3–11]). In [3] the optimisation of body-of-revolution (BoR) monopole antennas was explored, and they achieved a $VSWR \leq 3$ from 210 MHz up to at least 2 GHz at a height of 6 inches. A monocone was used in [4] to implement the monopole, and a top-cross-plate was used to lower the minimum operating frequency. The top cross plate was separated from the monocone by an air-gap, and the performance was quite sensitive to this separation distance. The design in [4] was able to achieve an impedance bandwidth from 3 to 12 GHz with an $|S_{11}| < -10$ dB and a height of $\lambda/11.5$ at the lowest operating frequency. A monocone was also used in [5] to realise a low-profile two-port omnidirectional antenna. In that work, a colocated antenna pair with a cone antenna in the middle and a ring radiator around the cone achieved an operational band from 800 to 2700 MHz ($|S_{11}| < -10$ dB) with dimensions $0.64\lambda \times 0.64\lambda \times 0.11\lambda$ at the lowest frequency.

An artificial magnetic conductor (AMC) reflector was used in [6] to design a low-profile dual-polarised wideband omnidirectional antenna. For vertical polarisation, [6] used a metallic cone-shaped monopole and a circular patch printed on an FR4 substrate shorted to the ground plane by four metallic pins which provided capacitive loading for achieving an impedance bandwidth of 1.7–2.7 GHz ($|S_{11}| < -10$ dB). Top-loading has been used in many other designs including in [7], which used a cascaded system of top hats, shorting strips and a ring slot cut into the ground plane to realise a composite reactive load that introduces additional resonances to reduce the lowest frequency of operation. Similarly, [8] used two stacked radiators with a larger one for the low-frequency band and a smaller one located within a region of low field strength. That design used a frequency-dependent feed network and was compact at $0.26\lambda \times 0.26\lambda \times 0.046\lambda$ achieving omnidirectional radiation patterns over roughly an 8.5:1 bandwidth (0.66–5.6 GHz) for a $VSWR < 3$. An extended version of a crossed-wire antenna was presented in [9] using crossed plates and achieved a $VSWR$ of < 2 from 2.26 to 15 GHz with a height of $\lambda/13.3$ at the lowest frequency with low cross-polarisation in the azimuth plane.

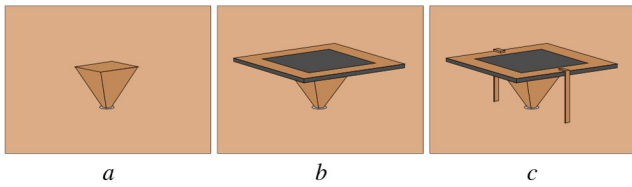


Fig. 1 Development of the proposed low-profile wideband antenna
(a) Pyramidal radiator, (b) Pyramidal radiator with top-loading, (c) Pyramidal radiator with top-loading and shorting strips

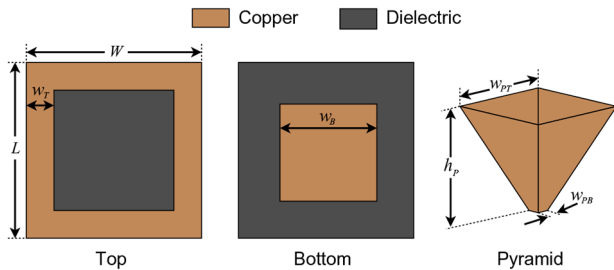


Fig. 2 Top and bottom of the etched copper-clad Rogers 5880 microwave board, and the pyramid radiator

Table 1 Top-loaded pyramidal antenna dimensions

Parameter	W	L	w_T	w_B	w_{PT}	w_{PB}	h_P
value, mm	50.0	50.0	8.5	25.3	24.0	3.0	19.4

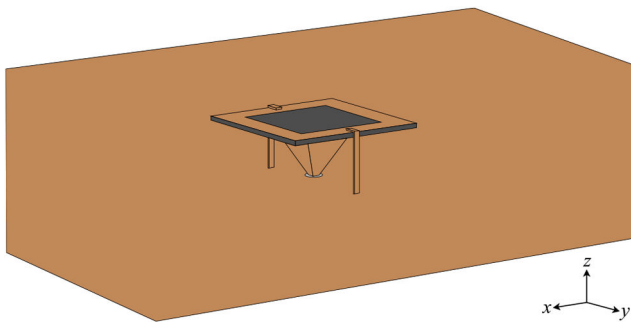


Fig. 3 Diagram of the proposed top-loaded pyramidal antenna on a flat ground plane

A number of antennas specifically designed for helmets have been presented in the literature (e.g. [12–22]). In [12], a low-profile antenna with a conical radiation pattern was realised by combining a four-element slot antenna array with an AMC surface, but this design had a limited bandwidth (25.6% for $|S_{11}| < -10$ dB). A spiral-mode microstrip antenna was mounted on the interior surface of a helmet in [13] which resulted in a VSWR ≤ 3 from 1.35 to 2.70 GHz. A similar design, but with a more broadband response was discussed in [14], but the azimuth radiation patterns became highly variable >3 GHz. An antenna array of four spiral-mode microstrip elements was conformally mounted on the curved surface of the helmet in [15]. The smart antenna array in that work used pattern diversity to mitigate the effects of multipath fading and the soldier's changing orientation but was not an omnidirectional solution. Another helmet antenna was presented in [16] which used conductive ink printing to realise antenna elements that are conformal to the helmet's shape. Such a technique, however, could likely not produce omnidirectional patterns over a broad bandwidth when actually implemented on a helmet and worn by a person, and the performance was limited to 3 GHz.

A number of VHF helmet antennas have been demonstrated [17–19]. These antennas, which are based on planar inverted-F antennas, planar inverse L-shape antennas, or folded dipoles, have a very narrowband response around 150 MHz and thus these designs are not suitable for the broadband applications considered in this paper. Characteristic mode analysis of the helmet was performed in [20], but this also will not provide a broadband

omnidirectional solution. A multiple band helmet antenna was detailed in [21] that could be mounted on the surface of a combat helmet. The design in that work combined a printed folded monopole antenna and a reverse L parasitic element, but was not very omnidirectional and only operated in particular fixed bands. The helmet antenna design in [22] used a BoR created from a third-order Bézier curve along with a top-load that had tapered slots to reduce the minimum frequency. As will be shown in this paper, the proposed antenna is simpler to fabricate, has smaller dimensions, and has a much broader overall frequency response than the design in [22].

Various aspects of the previous work discussed in this section were used to design the proposed broadband low-profile quasi-omnidirectional antenna specifically purposed for body-worn helmet applications. The proposed design, which was briefly introduced in [23] with a flat ground plane, uses a pyramidal radiator with a top-load and shorting strips to obtain broadband performance. Starting with a flat ground plane to describe the antenna's operation and the purpose of its various components, the design is then applied to a combat-style helmet where the outer surface of the helmet is used as the ground plane. To the best of our knowledge, this design represents the most broadband predominately omnidirectional helmet antenna presented in the literature. Furthermore, the design is very low-profile, relatively straightforward to fabricate, and the antenna could be retrofitted to existing helmets.

2 Antenna design

The stages of development for the proposed antenna are illustrated in Fig. 1. Fig. 1a shows the square pyramidal radiating element, which was selected for its inherent broadband properties and its ease of fabrication compared to more complicated structures such as a BoR. As discussed previously, top-loading can be used to reduce the lower frequency limit, and Fig. 1b shows the top-load element used in this work, which is a dielectric substrate with a square patch conductor on the bottom side and a square ring conductor on the top side. A diagram showing the top and bottom of the top-load, as well as the square pyramid is shown in Fig. 2. The top edge of the hollow conductive pyramid is soldered to the square patch conductor on the bottom side of the microwave board, and the antenna is fed at the bottom tip of the pyramid through the ground plane.

The separation between the conductors on the top and bottom of the microwave laminate control the capacitive loading on the pyramidal radiator (i.e. w_T , w_B , and the thickness of the substrate). Fig. 1c shows the complete antenna which includes two shorting strips that connect to the top side of the microwave board and the ground plane. These shorting strips are 2.5 mm wide and 0.6 mm thick, and the other dimensions of the antenna are provided in Table 1. The substrate used for the top-load is a Rogers RT/duroid 5880 substrate ($\epsilon_r = 2.2$) with a thickness of 1.575 mm. The method of moments-based computational electromagnetics package FEKO was used for the simulation results that follow.

The antenna was initially designed and optimised for a flat ground plane, as shown in Fig. 3, which includes the orientation of the coordinate system to be used in subsequent radiation pattern plots. It was desirable to optimise the design using an infinite flat ground plane approximation in the electromagnetic simulator to greatly decrease simulation time and then apply the design to a helmet antenna where the curved outer surface of the helmet acted as the ground plane.

To illustrate the effects of the top-loading and the shorting strips on the performance characteristics of the antenna, simulations were performed with each of the stages shown in Fig. 1 on a flat, infinite ground plane. Shown in Fig. 4 are the real and imaginary parts of the input impedance for each of the three antennas shown in Fig. 1. When comparing the pyramidal radiator alone with the pyramidal radiator that has top-loading, it is clear that there is a shift in resonance to a lower frequency, as expected. With the introduction of the shorting strips Fig. 4 shows an additional resonance at ~ 900 MHz, which serves to further lower the beginning of the operational frequency band. Corresponding to these input

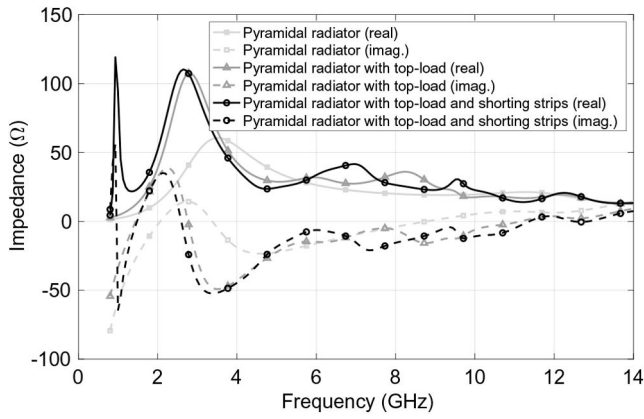


Fig. 4 Simulated input impedance of the three stages of the proposed antenna development shown in Fig. 1

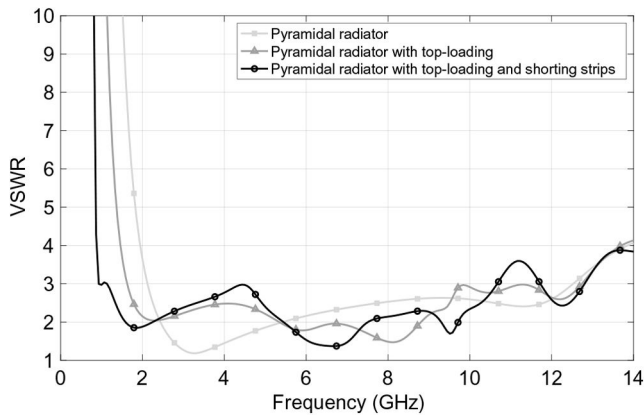


Fig. 5 Simulated VSWR of the three stages of the proposed antenna development shown in Fig. 1

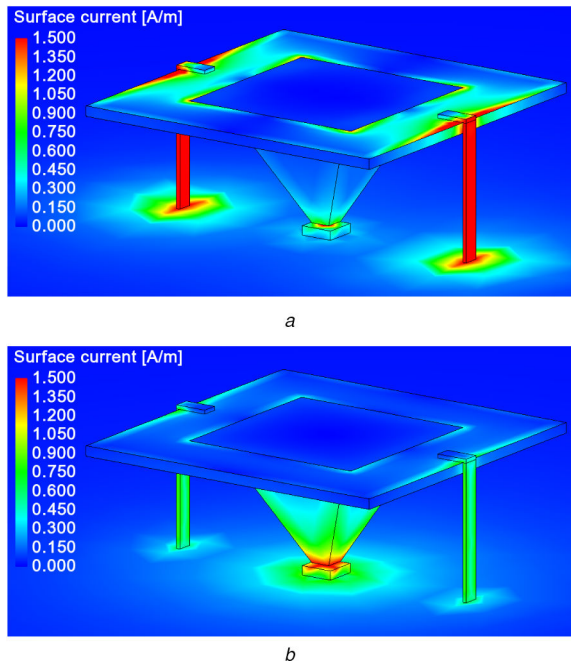


Fig. 6 Simulated current distribution of the proposed antenna (a) 1 GHz, (b) 2 GHz

impedance results, Fig. 5 shows the VSWR for each of the three antenna stages in Fig. 1. It is clear from this figure the contribution that each of the top-load and the shorting strips have on the VSWR at the lower end of the band. Specifically, the VSWR first goes < 3 at ~ 2.1 , 1.6 , and 0.9 GHz, respectively, for the three configurations in Fig. 1a–c.

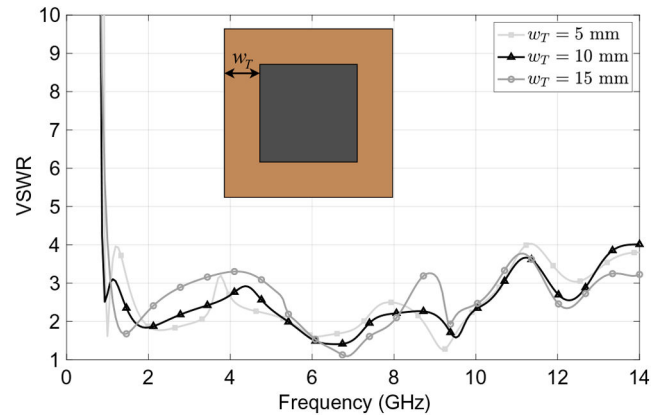


Fig. 7 Simulated VSWR showing the effect of the w_T parameter for a fixed bottom patch size ($w_B = 25.3$ mm)

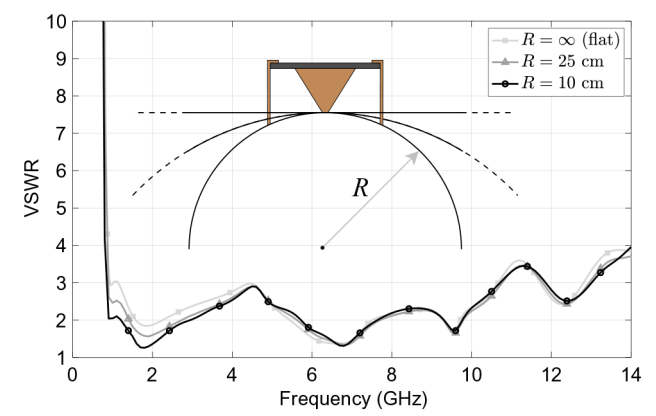


Fig. 8 Simulated effects of ground curvature on VSWR when the antenna is installed on top of a hemispherical PEC shell of radius R

The simulated current distributions at two different frequencies are shown in Fig. 6. From this figure, it is clear that the shorting strips play a much more significant role at the lower end of the band at around 1 GHz than at higher frequencies where the current is more concentrated on the pyramidal radiator. A consequence to the asymmetry introduced by these two shorting strips is that the radiation pattern becomes somewhat bidirectional at certain frequencies, as will be seen in subsequent results. The use of three or four strips was also explored, as well as varying their locations around the substrate. It was found that the use of two shorting strips centred along the sides of the substrate provided the lowest operating frequency for a $VSWR \leq 3$ and the widest bandwidth for a given antenna height.

To illustrate the effect of the overlap between the top and bottom conductors on VSWR, the width of the top conductor, w_T was varied while the size of the bottom square patch was fixed at $w_B = 25.3$ mm. The results, shown in Fig. 7, indicate that if the overlap is too large (e.g. $w_T = 15$ mm) the increased capacitance eliminates the effect of the low-frequency resonance around 900 MHz and the VSWR exceeds 3 at several frequencies throughout the band. If there is not enough overlap (e.g. $w_T = 5$ mm), the VSWR increases sharply after the low-frequency resonance above a value of 3. Using an intermediate value (e.g. $w_T = 10$ mm) takes advantage of the low frequency resonance while also maintaining a $VSWR \leq 3$. Additional simulations determined an optimal value of $w_T = 8.5$ mm, as indicated in Table 1.

Since the antenna was to be installed on top of a helmet where the shell of the helmet is ground, the effect of a curved ground plane on performance was explored. Shown in Fig. 8 are the VSWR results when the antenna is installed on top of a hemispherical perfect electrical conductor (PEC) shell of radius R . In this figure, three different radii are considered: $R = \infty$ (i.e. a flat ground plane), $R = 25$ cm, and $R = 10$ cm (similar to the size of a helmet). In each case, the only change to the geometry of the top-



Fig. 9 Simulation model and fabricated prototype of the proposed wideband top-loaded pyramidal antenna installed on a combat-style helmet (a) Side-view (left) and perspective-view (right) of the helmet antenna simulation model, (b) Photograph of the helmet antenna prototype without radome and cover (left), and with radome and camouflage cover (right)

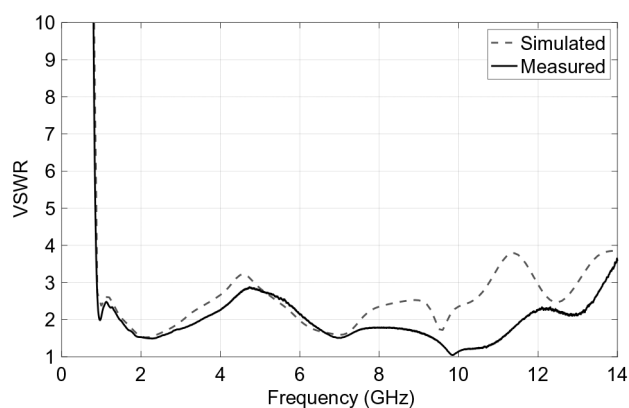


Fig. 10 Simulated and measured VSWR of the proposed antenna installed on a helmet

loaded pyramidal radiator was to extend the length of the shorting strips as necessary to reach the ground plane. The results of Fig. 8 show that the curvature of the ground does not have a significant impact and in fact, improves the VSWR marginally at low frequencies. These results also show that the design is not highly sensitive to the length of the shorting strips since they were varied for each case in Fig. 8 without a significant impact on performance.

3 Antenna fabrication and performance

3.1 Helmet antenna prototype

The square pyramidal radiating element was fabricated from a thin sheet of copper by cutting it to the appropriate shape and folding it manually to the required geometry. The shorting strips were also cut from a thin sheet of copper and the top-load was fabricated using an LPKF milling machine to remove the copper cladding on the Rogers 5880 microwave laminate wherever required according to Fig. 1.

To conduct simulations using the proposed antenna implemented on a helmet, a CAD model of a Personnel Armor System for Ground Troops (PASGT) combat helmet was used, as shown with the antenna installed in Fig. 9a. For simulations, the helmet material was set to be a PEC since the surface of the helmet is to be used as the ground plane. For fabrication, a PASGT-style combat helmet was also used and copper tape was applied to the outer surface to realise the ground plane. Unlike [22], where the inside surface of the helmet was used for the ground, this design used the outer surface to minimise the effects of the potentially

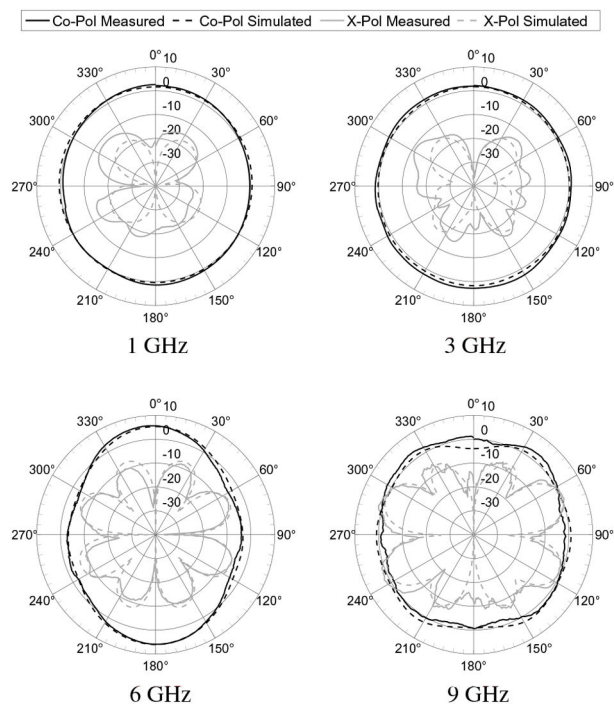


Fig. 11 Simulated and measured azimuth (xy-plane) radiation patterns of the proposed antenna on a helmet in dBi

lossy helmet material. A hole was drilled in the helmet and an SMA connector's ground and signal were soldered to the copper tape surface and to the tip of the square pyramidal radiator, respectively. An alternative installation, which could be non-destructive and potentially easier for retrofitting existing helmets would be to run a conformal microstrip transmission line down the back of the helmet where the connector could be located. A photograph of the prototype helmet antenna is shown in Fig. 9b on the left alongside the same prototype covered with a 3D printed radome and camouflage cover to emphasise how minimal the addition of this antenna would be to the existing form factor (note that all experimental results that follow were obtained without the radome and cover).

3.2 Helmet antenna performance

The simulated and measured VSWRs of the helmet antenna are shown in Fig. 10. The measured results show that the VSWR is <3 from 890 MHz to 13.7 GHz (over a 15:1 bandwidth). The azimuth radiation patterns are shown in Fig. 11 at four frequencies across the band. Omnidirectional patterns were obtained throughout the majority of the band, but become somewhat bidirectional around 6 GHz. However, the pattern becomes more uniform again after 8 GHz. Approximately omnidirectional patterns were obtained up to 11 GHz, beyond which nulls appeared in the pattern. Therefore, the usable operational band of the antenna is from 890 MHz to 11 GHz (a 12.4:1 bandwidth). The levels of cross-polarisation at some azimuth angles become significant at ~ 6 GHz. The importance of this cross-polarisation response at higher frequencies is application dependent and may not be a primary concern for applications where the intended target antenna's orientation is variable (such as in some EW scenarios). If a lower cross-polarisation response is desired throughout the band, techniques such as those shown in [9] could be used.

The gain in the azimuth xy-plane is shown in Fig. 12 and ranges from 2 to 6 dBi between 890 MHz and 11 GHz. The reduction of gain at some frequencies in Fig. 12, such as around 11 GHz, is caused by radiation patterns that are tilted somewhat upwards (the peak gain over all 3D space has less variability). The gain in the azimuth xy-plane is presumably more relevant than the overall gain for most helmet antenna applications assuming that signals are to be transmitted or received to/from other terrestrial sources as opposed to airborne ones. The antenna exhibits high simulated

efficiency of over 94% throughout the entire operational band, as shown in Fig. 12, due to the use of only copper material and a low-loss Rogers 5880 substrate. The measured xz - and yz -plane elevation patterns are shown in Fig. 13 at 2, 4, 6 and 8 GHz. Overall, the dimensions of the antenna not including the ground plane are $0.15\lambda \times 0.15\lambda \times 0.062\lambda$ at the lowest operating frequency of 890 MHz.

Compared to the helmet antenna design in [22], which achieved a $VSWR \leq 3$ from 750 MHz to 2.4 GHz (3.2:1), the proposed design achieves a much broader impedance bandwidth (over 15:1). Furthermore, the proposed design would be of lower cost to

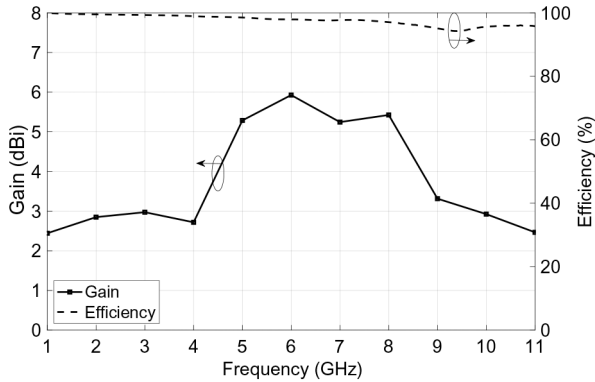


Fig. 12 Measured gain in the azimuth xy -plane and simulated efficiency of the helmet antenna

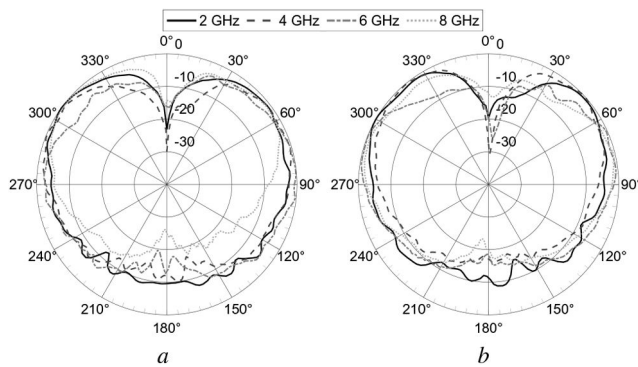


Fig. 13 Measured normalised elevation radiation patterns of the proposed antenna on a helmet
(a) xz -plane, (b) yz -plane

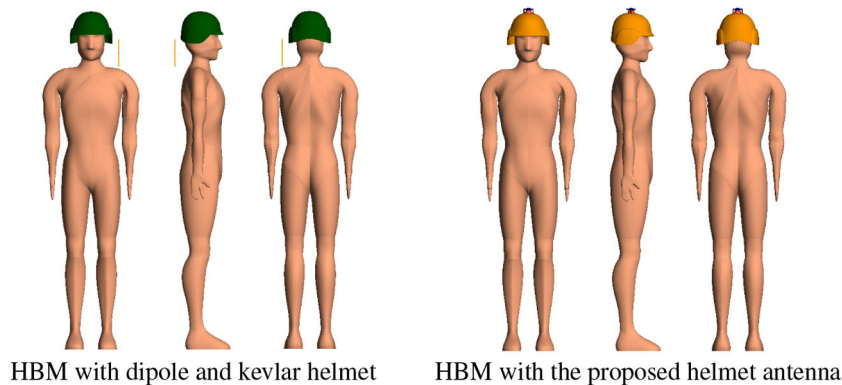


Fig. 14 Simulation models used to compare the omnidirectionality of a $\lambda/2$ dipole behind/above the shoulder (left) with the proposed helmet antenna (right)

Table 2 HBM tissue dielectric parameters [24]

	Average brain		Average muscle	
	ϵ_r	σ , S/m	ϵ_r	σ , S/m
1 GHz	45.43	0.80	55.74	1.01
2 GHz	43.21	1.26	54.17	1.51
3 GHz	41.79	1.86	52.85	2.24

fabricate owing to the use of a simple pyramidal radiator as opposed to a more complicated BoR created from a third-order Bézier curve (e.g. CNC machining is not required). The lateral dimensions of the proposed antenna are also smaller than the design in [22] (0.15λ versus 0.21λ at the lowest frequencies). The body-worn antenna in [10] has a similar height to the proposed antenna (0.07λ), but significantly larger lateral dimensions (0.64λ). Compared to the wide body of work on low-profile monopole antennas (e.g. [3–11]), the combination of electrical size and bandwidth of the proposed helmet antenna is amongst the best, though direct comparisons are difficult to make due to the curved ground plane of the helmet and different VSWR design goals (e.g. see comparison tables in [6–8, 22]).

3.3 Helmet antenna with a human body model (HBM)

In some military applications, dismounted soldiers require antennas with ideally omnidirectional radiation patterns. The need for freedom of movement generally necessitates body-worn antennas that are in close proximity to the human body. For the specific case of ECM, this can involve wearing a backpack containing the electronics and one or more antennas protruding vertically behind the soldier's shoulder and head. Omnidirectionality is particularly important for ECM applications to provide uniform spatial jamming protection against RCIEDs for dismounted soldiers. Conventional systems that rely on monopoles and dipoles located behind the shoulder/head can also be cumbersome and limit agility and mobility. To provide an approximate quantitative comparison between the conventional antenna solution for this application and the use of the proposed helmet antenna, simulations were performed using a simple HBM.

Shown in Fig. 14 are the HBMs used along with a more conventional $\lambda/2$ dipole antenna located behind the shoulder/head (left) and the proposed helmet antenna (right). The simple HBM consists of two parts: the head is a uniform dielectric with the electrical properties of average brain tissue, and the rest of the body has the properties of average muscle. These frequency-dependent properties were obtained from [24] and are summarised for the three considered frequencies in Table 2. The model of a combat helmet is included for both the dipole and the proposed helmet antenna. In the case of the dipole, the material properties of the helmet are set to that of kevlar with $\epsilon_r = 3.7$, $\tan \delta = 0.04$, and in the case of the helmet antenna, its surface is a PEC.

Simulation results of the azimuth (xy -plane) radiation patterns are shown at 1, 2, and 3 GHz in Fig. 15. From this figure it is clear that the use of the proposed helmet antenna significantly improves omnidirectionality. For the dipole, the HBM causes a significant

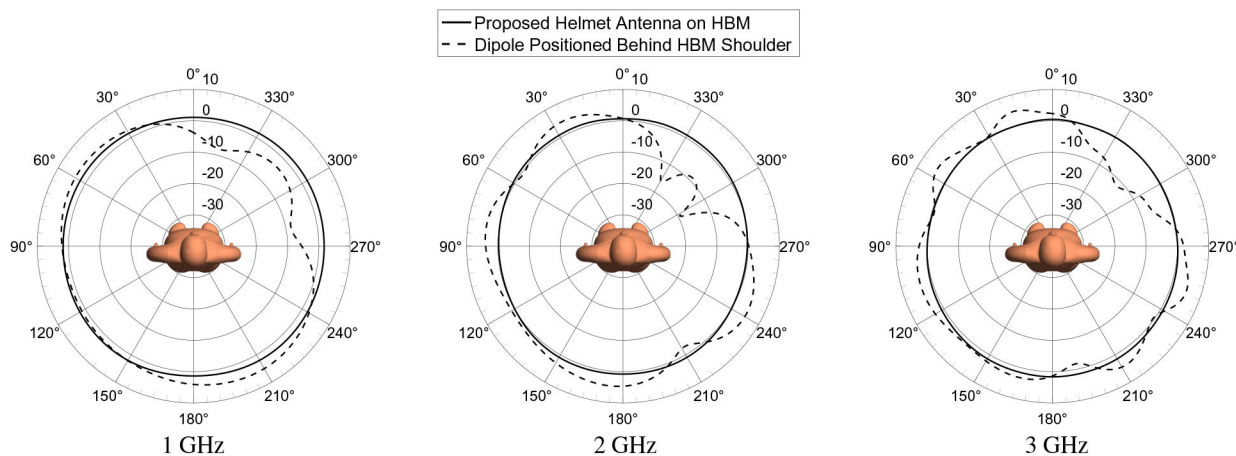


Fig. 15 Simulated azimuth (xy -plane) radiation patterns using a HBM with the proposed antenna installed on a helmet and half-wavelength dipoles located behind and above the model's shoulder in dBi (overhead HBM shown facing $\phi = 0^\circ$ to indicate orientation)

reduction in gain toward the side opposite of where the antenna is located, as would be expected intuitively. At 1 GHz, the variation in azimuth pattern (i.e. the difference between maximum gain and minimum gain in the xy -plane) for the dipole is over 13.5 dB compared to 1.0 dB for the proposed helmet antenna. At 2 and 3 GHz the variation in azimuth pattern for the dipole case are 24.4 and 17.0 dB, respectively, compared to 1.4 and 2.0 dB for the proposed helmet antenna. In particular, at 2 GHz, the gain of the dipole is reduced to as little as -20 dBi at some angles, which could have a significant impact on performance in an EW or communications application. As expected, the proposed helmet antenna shows very limited effects from the HBM.

If used for a transmitting application, the specific absorption rate (SAR) would need to be evaluated to ensure the safety of the wearer. Although the antenna is located close to the head in the proposed design compared to the more conventional whip/dipole approach, the head is shielded by the conductive ground plane surface, and as such may result in a reduced SAR. Accurate evaluation of the SAR would depend on the specific implementation of the helmet antenna including any holes that are drilled through the helmet, the thickness of the conductor used to realise the ground plane on the surface of the helmet, and the transmitter power level required for the particular application.

4 Conclusion

A low-profile and wide bandwidth body-worn antenna suitable for installation on a helmet has been presented using a square pyramidal radiator with top-loading and two shorting strips. The stages of development of the antenna were detailed using simulation results to illustrate the effect of the top-load and shorting strips on the VSWR, and a prototype was fabricated for the proposed application as a helmet antenna. The experimental results on the helmet prototype demonstrate a very broad VSWR response from 890 MHz to 13.7 GHz ($VSWR \leq 3$), and predominately omnidirectional radiation patterns up to 11 GHz (a 12.4:1 bandwidth). The proposed antenna is very low-profile at $\lambda/16$ for the lowest frequency and is also light-weight, low-cost and easy to fabricate. The use of the proposed helmet antenna could potentially eliminate physically cumbersome whips/dipoles located near the body for emergency services and military applications. Simulation results using a simple HBM demonstrate that the omnidirectionality of the proposed helmet antenna is significantly improved compared to the more conventional body-worn antennas used for some applications.

5 References

- [1] Abedian, M., Oraizi, H., Abdul Rahim, S.K., *et al.*: 'Wideband rectangular dielectric resonator antenna for low-profile applications', *IET Microw. Antennas Propag.*, 2018, **12**, (1), pp. 115–119
- [2] Alibakhshikenari, M., Virdee, B.S., See, C.H., *et al.*: 'Wideband printed monopole antenna for application in wireless communication systems', *IET Microw. Antennas Propag.*, 2018, **12**, (7), pp. 1222–1230

- [3] Zhao, J., Psychoudakis, D., Chen, C., *et al.*: 'Design optimization of a low-profile UWB body-of-revolution monopole antenna', *IEEE Trans. Antennas Propag.*, 2012, **60**, (12), pp. 5578–5586
- [4] Koohestani, M., Zürcher, J.F., Moreira, A.A., *et al.*: 'A novel, low-profile, vertically-polarized UWB antenna for WBAN', *IEEE Trans. Antennas Propag.*, 2014, **62**, (4), pp. 1888–1894
- [5] Yetisir, E., Chen, C., Volakis, J.L.: 'Wideband low profile multiport antenna with omnidirectional pattern and high isolation', *IEEE Trans. Antennas Propag.*, 2016, **64**, (9), pp. 3777–3786
- [6] Wu, J., Yang, S., Chen, Y., *et al.*: 'A low profile dual-polarized wideband omnidirectional antenna based on AMC reflector', *IEEE Trans. Antennas Propag.*, 2017, **65**, (1), pp. 368–374
- [7] Abadi, S.M.A.M.H., Behdad, N.: 'An electrically small, vertically polarized ultrawideband antenna with monopole-like radiation characteristics', *IEEE Antennas Wirel. Propag. Lett.*, 2014, **13**, pp. 742–745
- [8] Ghaemi, K., Behdad, N.: 'A low-profile, vertically polarized ultrawideband antenna with monopole-like radiation characteristics', *IEEE Trans. Antennas Propag.*, 2015, **63**, (8), pp. 3699–3705
- [9] Nakano, H., Takeuchi, M., Yamauchi, J.: 'Low-profile wideband iCROSS antenna', *IEEE Trans. Antennas Propag.*, 2016, **64**, (7), pp. 2824–2831
- [10] Jeong, W., Tak, J., Choi, J.: 'A low-profile IR-UWB antenna with ring patch for WBAN applications', *IEEE Antennas Wirel. Propag. Lett.*, 2015, **14**, pp. 1447–1450
- [11] Aten, D.W., Haupt, R.L.: 'A wideband, low profile, shorted top hat monocone antenna', *IEEE Trans. Antennas Propag.*, 2012, **60**, (10), pp. 4485–4491
- [12] Cao, Y.F., Zhang, X.Y., Mo, T.: 'Low-profile conical-pattern slot antenna with wideband performance using artificial magnetic conductors', *IEEE Trans. Antennas Propag.*, 2018, **66**, (5), pp. 2210–2218
- [13] Wang, J.J.H.: 'Broadband omnidirectional helmet antennas'. IEEE Antennas and Propagation Society Int. Symp., Albuquerque, NM, USA, July 2006, pp. 2129–2132
- [14] Wang, J.J.H., Triplett, D.J.: 'Multioctave broadband body-wearable helmet and vest antennas'. IEEE Antennas and Propagation Society Int. Symp., Honolulu, HI, USA, June 2007, pp. 4172–4175
- [15] Tillery, J.K., Thompson, G.T., Wang, J.J.H.: 'Low-power low-profile multifunction helmet-mounted smart array antenna'. IEEE Antennas and Propagation Society Int. Symp., Orlando, FL, USA, July 1999, 1554–1557
- [16] Herold, D., Griffiths, L., Fung, T.Y.: 'Lightweight, high-bandwidth conformal antenna system for ballistic helmets'. MILCOM - IEEE Military Communications Conf., Orlando, FL, USA, October 2007, pp. 1–6
- [17] Ono, Y., Okano, Y.: 'Development of VHF-band antenna mounted on the helmet'. Int. Symp. on Antennas and Propagation, Nanjing, China, October 2013, pp. 584–587
- [18] Nishiyama, N., Michishita, N., Morishita, H.: 'Low-frequency inverted-F antenna on annular ground plane'. IEEE MTT-S Int. Microwave Workshop Series on RF and Wireless Technologies for Biomedical and Healthcare Applications (IMWS-BIO), Taipei, Taiwan, September 2015, pp. 143–144
- [19] Morishita, H., Michishita, N.: 'Characteristics of helmet antennas at VHF band'. IEEE Conf. on Antenna Measurements Applications, Vasteras, Sweden, September 2018, pp. 1–2
- [20] Nomura, N., Michishita, N., Morishita, H.: 'Characteristic mode analysis of hemispherical shell for helmet antenna design'. Int. Symp. on Antennas and Propagation (ISAP), Okinawa, Japan, October 2016, pp. 238–239
- [21] Shin, J., Woo, J.: 'Military antennas'. Int. Symp. on Antennas and Propagation (ISAP), Busan, Korea, October 2018, pp. 1–2
- [22] Nguyen-Trong, N., Piotrowski, A., Kaufmann, T., *et al.*: 'Low-profile wideband monopolar UHF antennas for integration onto vehicles and helmets', *IEEE Trans. Antennas Propag.*, 2016, **64**, (6), pp. 2562–2568
- [23] Jackson, B.R., Merkley, E.: 'A wide bandwidth low-profile top-loaded pyramidal antenna'. 2018 IEEE Int. Symp. on Antennas and Propagation USNC/URSI National Radio Science Meeting, Boston, MA, USA, July 2018, pp. 1351–1352
- [24] Gabriel, C.: 'Compilation of the dielectric properties of body tissues at RF and microwave frequencies'. AFOSR-TR-96-0135, US Air Force Office of Scientific Research, 1996